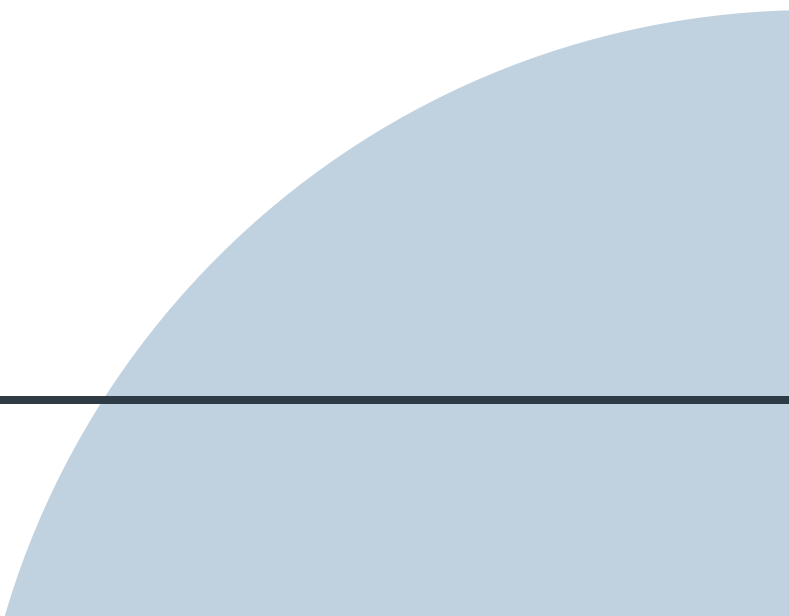




Adaptive Arena Sprint 2



Meet The Team



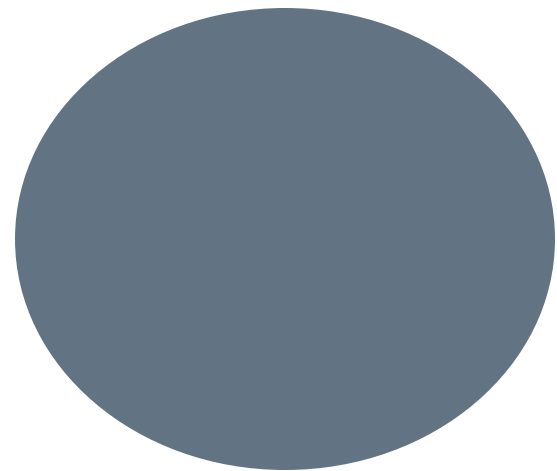
Inna Ivashchenko
Project Manager / Developer



Johnhenry Brayman
Lead Game Developer



Shariah Allen
Game Developer / AI Engineer



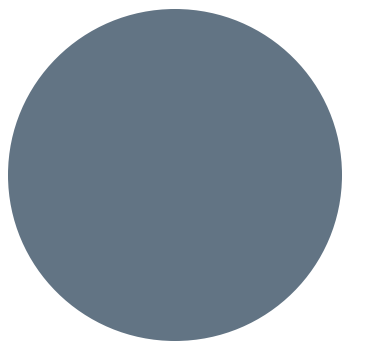
Anthony Piombino
Project Manager / QA Engineer



Gabrielle Schwan
UI/UX Developer



Henry Wong
AI/ML Advisor / Consultant





Recap sprint 1 / discuss improvements made



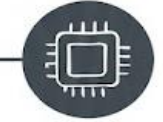
Project Description



Team Working Agreement



User Personas



MVP / Tech / Algorithms



Diagrams



Product Backlog / User Stories



Team Velocity / Burndown



Demo

Agenda

Sprint 1 Recap

- Execute concept developed in sprint 0
 - Minimal graphics and no AI
- Fully controllable player character
 - Establish move set of punch, block, double jump
 - Bug testing and resolution
- Damage tracking
- Camera tracking

Improvements made from Advisor Feedback

- Enhancement to presentation slides
- Adding advisor to team slide



Project Description

- 2D-fighting game focusing primarily on AI opponents and improving your general fighting game skills.
- AI will scale appropriately based on selected skill level
 - Easy : Mostly random or suboptimal moves
 - Medium : Basic strategy with a little randomness
 - Hard : Uses machine learning to adapt and respond
- Provide user-friendly interface for user entry
- Tally up and output results of game



Team Working Agreement

Communication And Responsibilities:

- Communication through WhatsApp, Slack, Zoom.
- All team members must respect each other and have the right to contribute their ideas to the project.
- All ideas, vision and changes discussed together, and final decisions are up to most of the team, except urgent situations.
- Each person works within the boundaries established before the start of work.
- Everyone is responsible for their own part.
- All conflicts should first be resolved between the people involved. If that is not possible the issue should be brought to the Team Lead.
- Each member must inform the team about any changes or improvements they make.

Meetings:

- Meeting happens through Zoom, Slack
- The team must have Scrum meetings two times in a week and check WhatsApp, Slack daily to avoid missing any project information.
- Schedule of a Scrum meeting : Tuesday at 8pm and Friday at 9.15pm
- Each team member should participate in at least one meeting per week unless they have an important reason.

User Personas: The Competitive Gamer

Name: Anna Grace

Age: 28

Gender: Female

Occupation: Game Developer & Twitch Streamer

Anna is a season game developer with a solid fan base, that enjoys playing games in her free time, specifically fighting, first-person shooters (FPS), and racing games. She's often searching for new and exciting games that have varying difficult levels and unique features that keeps players engaged.

Challenges:

- Tired of playing the same popular titles, with reoccurring characters and stories
- Wants to find new games developed by small teams to stream for her community

Goals:

- She enjoys playing games and learning different mechanics
- Wants to explore new games with AI elements
- Seeks a game that can be replayed several times

Quote:

“ I just want to entertain my community and learn new skills; I love a good challenge!”



User Personas: The Button Smasher

Name: Jordan Carter

Age: 19

Gender: Male

Occupation: Undergraduate Student and Student Athlete

Jordan is a busy student athlete studying animation, that loves to play games on his free time on the weekends or while hanging out with friends. He loves playing games with lots of action, unique characters, and prefers to not memorize combos but just plays for fun, not to win.

Challenges:

- Complex inputs and combinations makes gameplay frustrating
- Long or vague tutorials

Goals:

- Have fun solo or with friends
- Win occasionally, or access to varying difficulties

Quote:

“ I skip tutorials, I just want to have mindless fun!”



User Personas: The Lore Explorer

Name: Dylan Taylor

Age: 12

Gender: Male

Occupation: Middle School Student

Dylan enjoys playing games after school with friends, family, or by himself, and specifically enjoys games that are action-packed. He seeks out games that provide a good challenge, has lots of enemies, and visual stimulation. He aspires to be a game developer and enjoys learning how to make 2D games during his video game club meetings.

Challenges:

- Hard to find action games that are age appropriate
- Wants games with unique enemies
- Prefers 2D games

Goals:

- Wants to explore new action games
- Age-appropriate game with visual stimulation

Quote:

“ Games are fun when they have a lot of action!”



Inputs

Horizontal Movement:

- Move Forward: Player advances toward the opponent in the right direction using the "D" or "Right" arrow keys
- Move Backward: Player retreats from opponent towards the left direction using the "A" or "Left" arrow keys

Blocking:

- Defensive action intended to reduce or prevent incoming damage from attacks
- Use the Right Mouse Button (RMB) to block attacks

Vertical Movement:

- Player can do a single or double jump mid-air to evade their opponent's attacks
- Uses the spacebar to do single or double jumps
 - Double tap the spacebar to do a double jump

Attacks: Punch

- Basic melee attack that deals damage when the player's hitbox collides with the opponent
- Use the Left Mouse Button (LMB) to throw a punch

Reactive Actions

Being Hit

- Triggered once an attack connects.
- Character takes damage
- May briefly enter a hit reaction state once hit
 - Each attack deals varying levels of damage
 - Ex. Punch deals 10 damage

Dying

- Occurs once a fighter's health reaches 0
- Ends Match
- Game transitions to End-Game State

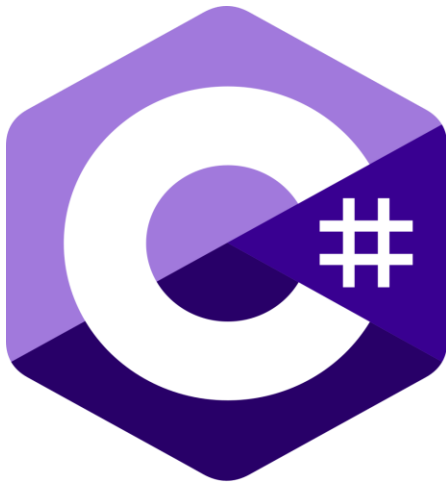
Technologies



Unity 6.3 LTS



Git LFS Extension



C# Programming



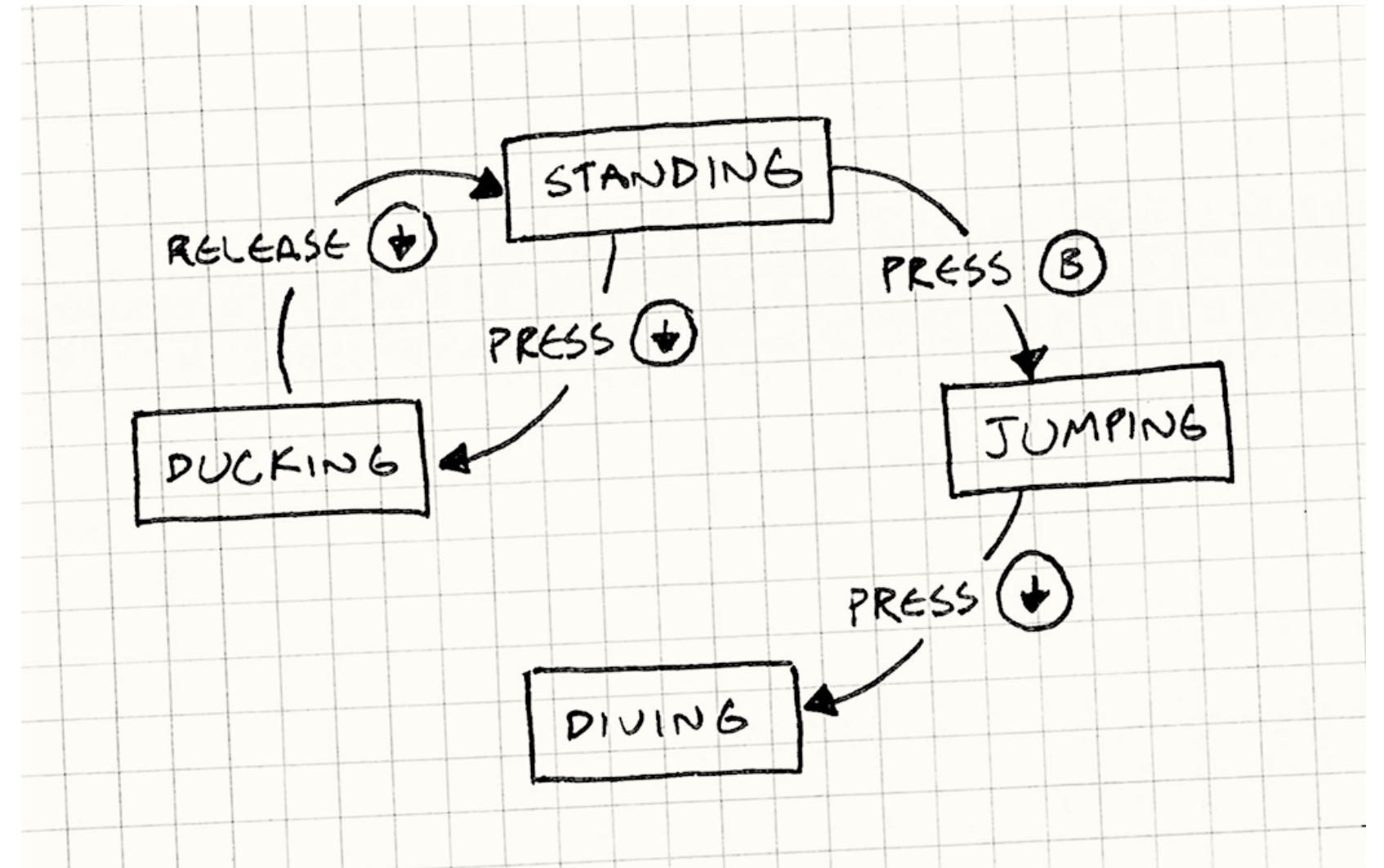
Python (for Machine Learning)



Algorithms

Easy Difficulty : Randomized Finite State Machine

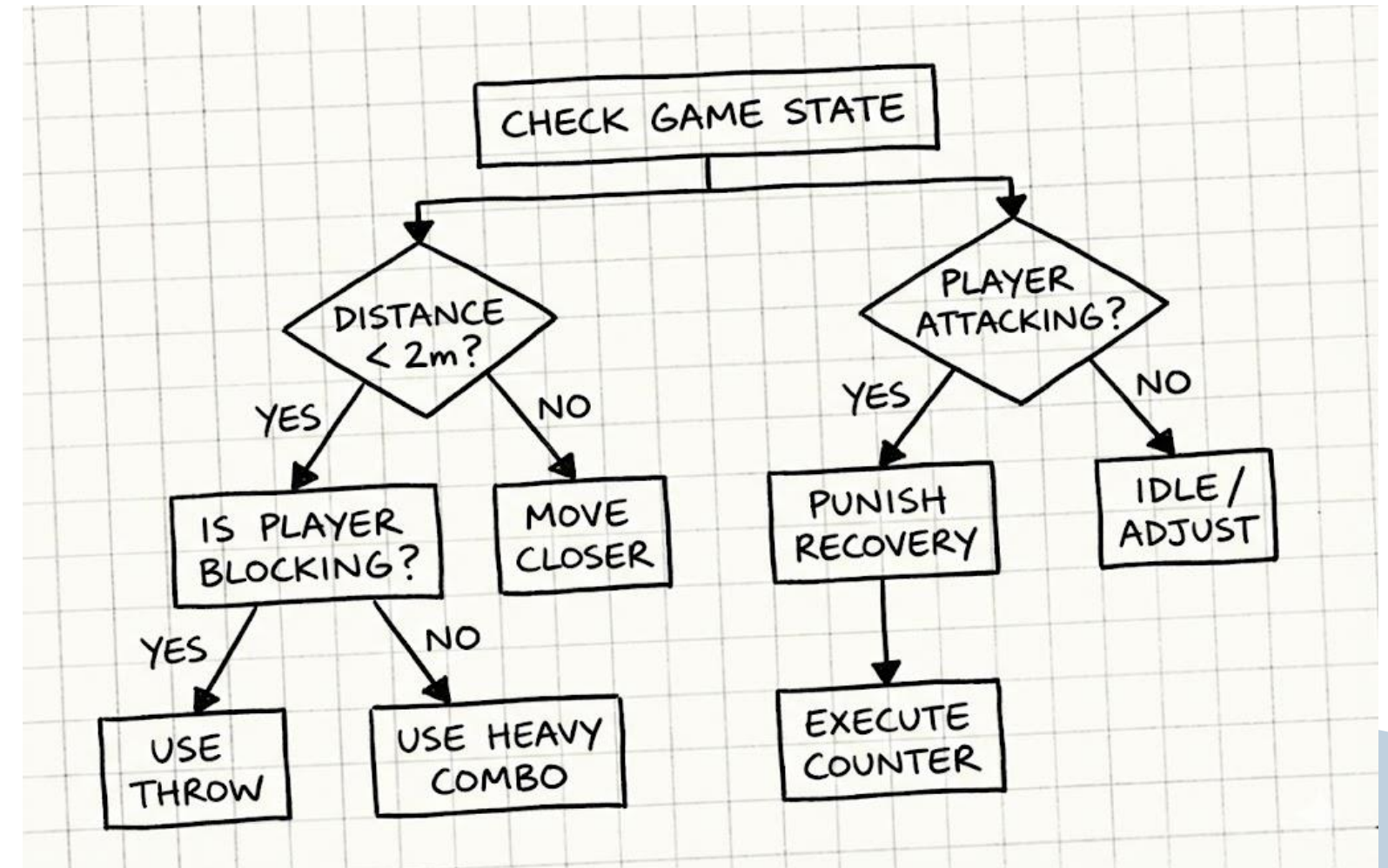
- The enemy AI is presented with multiple pre-defined states (Idle, Attack, Advance, Block, Duck).
- Instead of using strategy, it picks which state to move to next randomly.
- Attacks and movements still make sense but are suboptimal.



Algorithms

Medium Difficulty : Decision Trees

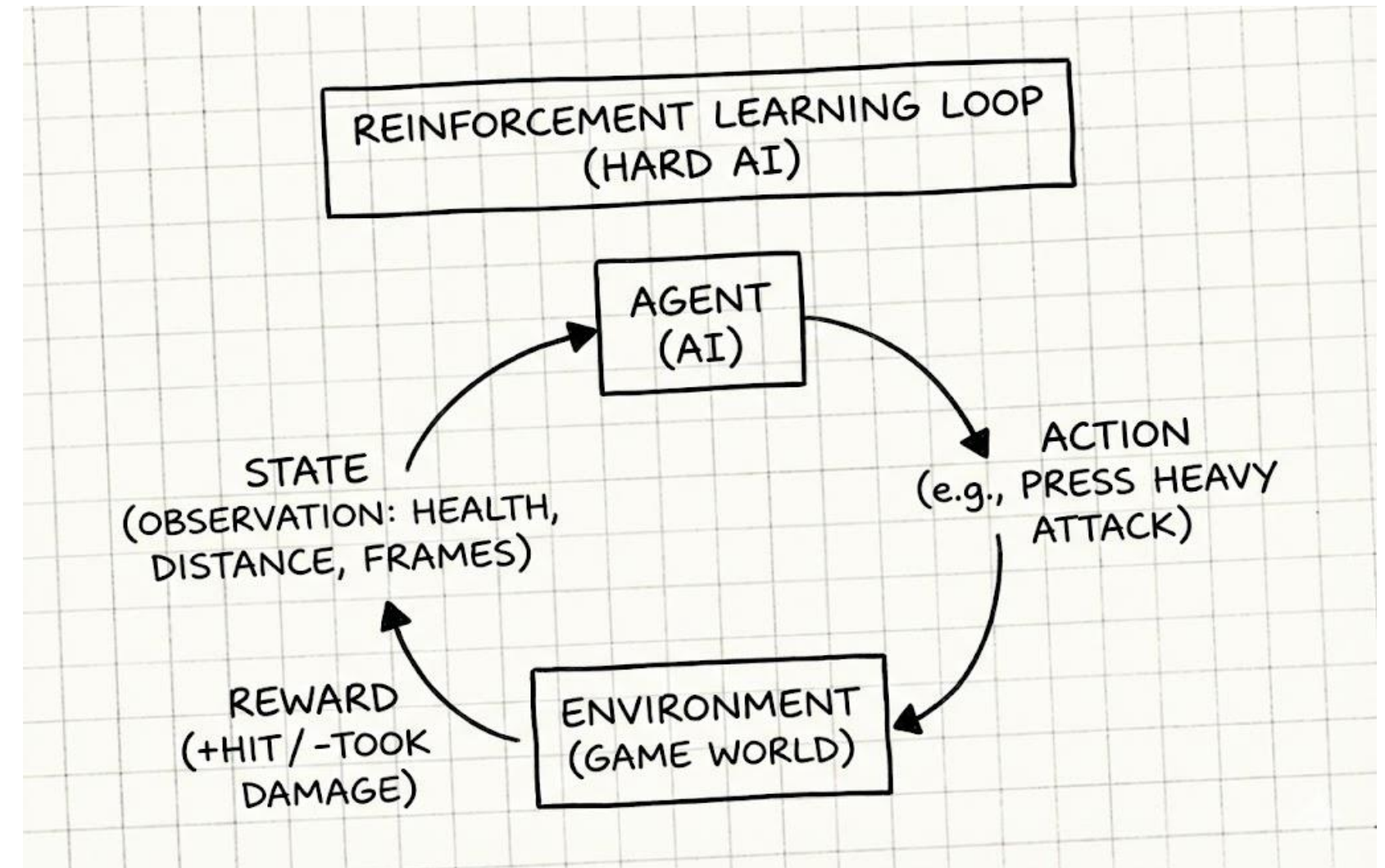
- The enemy AI in this difficulty will start to react differently depending on certain criteria.
- Before making an action, it will check a specific condition and act accordingly.
- Can still make random decisions but in a more educated way.
 - Ex. Check to see if the player is attacking. If yes, randomly choose between block or counter.



Algorithms

Hard Difficulty : Reinforcement Learning

- The enemy AI will fight through many simulations either against itself, or one of the before mentioned difficulties.
- It will receive a reward (+1) for landing a hit on the player, and a punishment (-1) for being struck by the player.
- The AI will learn which actions lead to rewards and which actions lead to punishments.



Minimum Viable Product (MVP)

The Environment:

- A simple flat ground with invisible walls on each side to prevent falling off.
- Simple environment design keeps focus on combat mechanics and player interaction
- A basic camera with a script that keeps both players in-frame while zooming in as close as possible.

The AI Difficulties:

- For the rules-based difficulties, the State Machine and Decision Tree logic must work, even if they are not complex.
- For the hard difficulty, the Unity Agent must be able to read observations and output actions from the Neural Net.

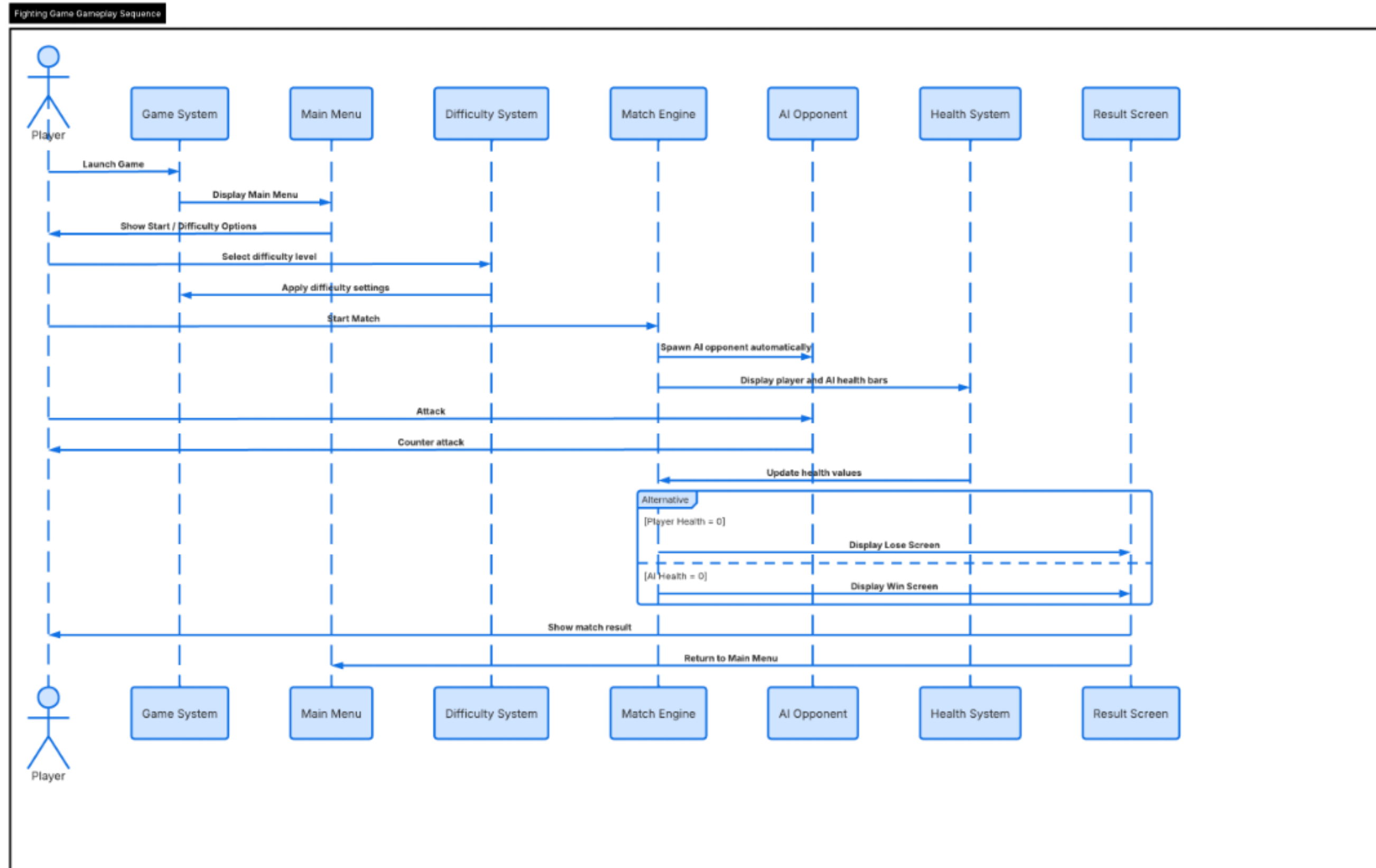
The Combat System:

- The bare minimum controls will be Locomotion, an Attack action, and a Block action.
- Hit detection uses colliders to determine when attacks connect
- Possibility to include combos, ducking, countering, and throwing later.

The Game Flow:

- The player must be able to enter the game from a main menu and select an AI difficulty
- Loads varying end-game scenes depending on match outcome
- In-game must show health bars depleting, and a return to the main menu when one character reaches zero.

Sequence Diagram



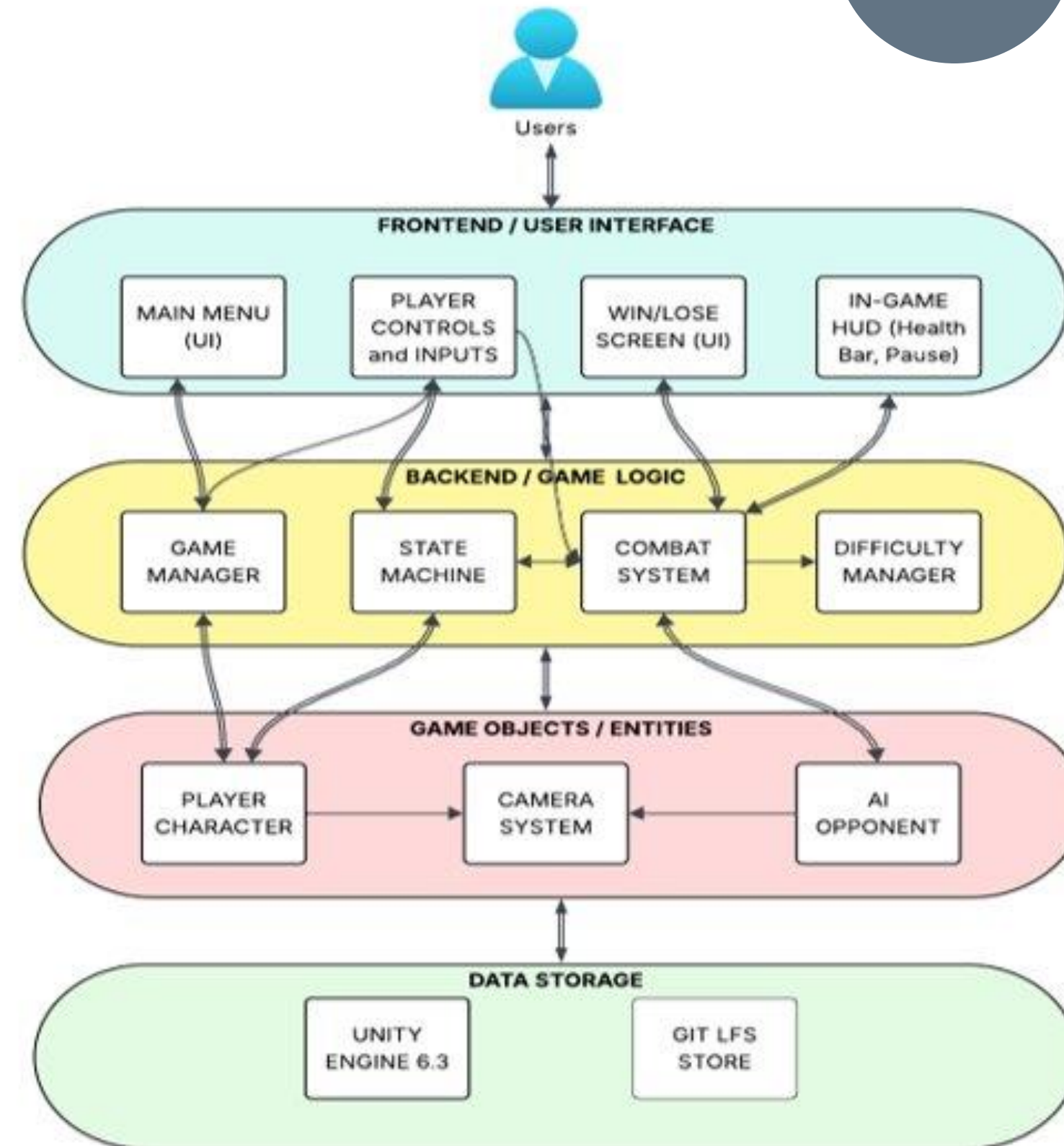
Conceptual Architecture Diagram

Frontend: Accepts player input and displays health status (HUD), Win/Lose Screen.

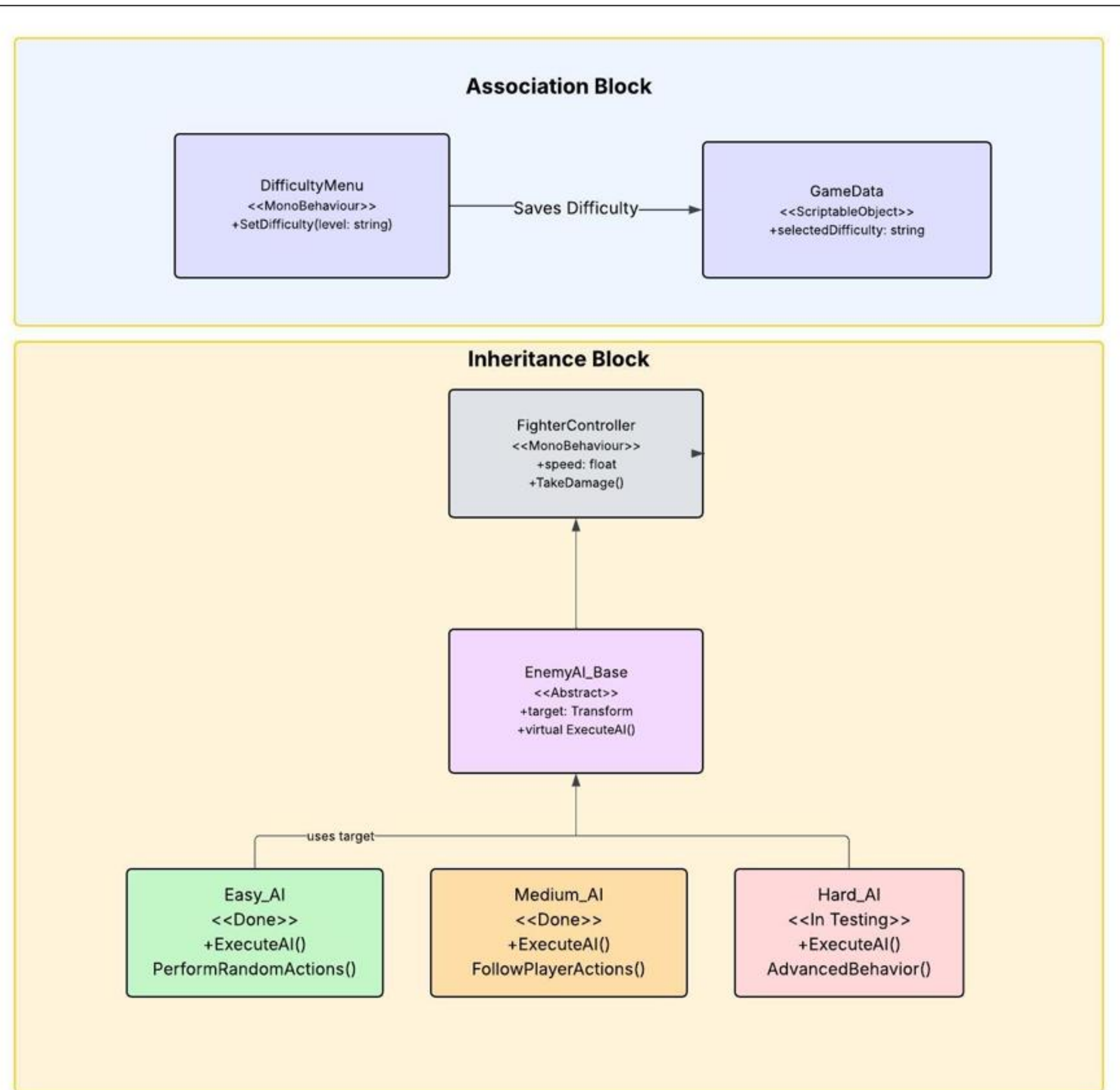
Backend: The "brain" of the game, managing combat logic, character states, and difficulty.

Entities: Game characters and a "smart" camera, operating according to the backend's rules.

Data Storage: A stable foundation built on Unity 6.3 and Git LFS storage.



Class Diagram (UML Diagram)



Product Backlog

No.	User Stories	Acceptance Criteria	Feature
US_1	As a player, I want to access a main menu so that I can make choices in the game.	Main menu displays on game launch. Menu includes: Start, Settings, Quit, Credits. User can navigate menu using mouse. Selecting Start transitions to fight screen. Selecting Exit closes the game	Game Navigation
US_2	As a player, I want to design my character so that I can play with a customized fighter.	Player can choose character name. Player can design character appearance. Character preview is displayed before confirmation. User confirms selection to proceed. Selected character loads into the match	Character System
US_3	As a player, I want to be able to challenge an easy AI opponent so that I can practice basic gameplay mechanics.	An EasyController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack at random. The AI moves in the general direction of the player. The AI has a slower reaction update time, making its behavior less responsive and easier to predict.	Gameplay
US_4	As a player, I want to be able to challenge a medium AI opponent so that I can experience a balanced level of difficulty.	A MediumController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack with more consistency than random behavior. The AI moves toward the player more effectively than the easy AI. The AI has a moderate reaction update time, making it more responsive than easy but not overly difficult.	Gameplay
US_5	As a player, I want to be able to challenge a hard AI opponent so that I can face a highly challenging and intelligent opponent.	A HardController script is implemented using machine learning techniques. Unity ML-Agents are installed and configured with a working Python environment. The AI overrides CollectObservations, OnActionReceived, and OnEpisodeBegin methods. A reward and penalty system is implemented to guide AI learning. The AI is trained using a Python training process. The resulting trained neural network model is imported and used in the game. The hard AI demonstrates advanced and responsive behavior compared to easy and medium difficulties.	Gameplay
US_6	As a player, I want the player and AI characters to use pixel art sprites so that the characters match the retro visual style.	Player character uses a pixel art sprite. AI opponent uses a pixel art sprite. Sprites replace any placeholder shapes for characters and display correctly during gameplay.	Visual Design
US_7	As a player, I want to see a health bar during the fight so that I can track progress.	Player health bar is visible on screen during the match. AI health bar is also visible. Health bars decrease when damage is taken. Health bars update in real time during combat.	Combat UI
US_8	As a player, I want to win or lose based on health depletion so that the fight has a clear outcome.	When the AI's health reaches zero, the player wins. When the player's health reaches zero, the player loses. The match ends immediately when one health bar reaches zero. No further movement or attacks occur after the match ends.	Combat System
US_9	As a player, I want to see a win or lose screen so that I know the match outcome.	A win screen displays when the player wins. A lose screen displays when the player loses. End screen includes options to return to the main menu or play again. Selecting an option transitions properly without crashing the game.	Game Navigation

Product Backlog Continued

No.	User Stories	Acceptance Criteria	Feature
US_10	As a player, I want smooth transitions between game stages so that gameplay feels complete and polished.	Game transitions smoothly from main menu to character creation, then to difficulty selection, and then to the match. Selected character and difficulty settings carry into the fight. After the match ends, the player can return to the main menu or restart the game without errors or crashes.	Game Architecture
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	Camera System
US_12	As a player, I want the game environment to use pixel art graphics so that the stage matches the retro aesthetic.	The game background uses pixel art design. Background assets load correctly during gameplay. The environment visuals are consistent with the pixel art style and no placeholder backgrounds appear in the Sprint demo.	Visual Design
US_13	As a player, I want character sprites to animate during movement and attacks so that the game feels dynamic and visually engaging.	Player sprites animate during movement and attacks. AI sprites animate during movement or attacks. Animations display smoothly during gameplay.	Visual Design
US_14	As a player, I want the ability to block attacks so that I can defend myself against enemy attacks during combat.	The player can activate a block action using a designated input. When blocking, incoming attacks deal reduced or no damage. A blocking animation or visual indicator appears while the player is blocking. Blocking only works while the block input is active.	Gameplay
US_15	As a player, I want the game to prevent my character from falling out of the environment so that gameplay stays within the intended boundaries.	The player character cannot move beyond the stage boundaries. The player cannot fall through the ground or platforms. Collision detection prevents the player from leaving the playable area. The player remains within the environment during gameplay.	Gameplay
US_16	As a player, I want the ability to perform a double jump so that I can reach higher areas and have greater mobility during gameplay.	The player can jump once from the ground and perform a second jump while in the air. The second jump can only occur after the first jump. The player cannot perform more than two jumps before landing. The double jump resets after the player lands on the ground.	Gameplay
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	Gameplay

Sprint 2 Stories

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint	Team Member
US_3	As a player, I want to be able to challenge an easy AI opponent so that I can practice basic gameplay mechanics.	An EasyController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack at random. The AI moves in the general direction of the player. The AI has a slower reaction update time, making its behavior less responsive and easier to predict.	Gameplay	10	1	2	Inna
US_4	As a player, I want to be able to challenge a medium AI opponent so that I can experience a balanced level of difficulty.	A MediumController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack with more consistency than random behavior. The AI moves toward the player more effectively than the easy AI. The AI has a moderate reaction update time, making it more responsive than easy but not overly difficult.	Gameplay	10	1	2	Shariah / Gabrielle
US_5	As a player, I want to be able to challenge a hard AI opponent so that I can face a highly challenging and intelligent opponent.	A HardController script is implemented using machine learning techniques. Unity ML-Agents are installed and configured with a working Python environment. The AI overrides CollectObservations, OnActionReceived, and OnEpisodeBegin methods. A reward and penalty system is implemented to guide AI learning. The AI is trained using a Python training process. The resulting trained neural network model is imported and used in the game. The hard AI demonstrates advanced and responsive behavior compared to easy and medium difficulties.	Gameplay	15	1	2	Johnny

Sprint 2 Stories Continued

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint	Team Member
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	Camera System	6 (2 already earned in sprint 1)	3	2	Anthony
US_14	As a player, I want to be able to select between multiple difficulties so that I can choose a gameplay experience that matches my skill level.	The game provides an option to select between easy, medium, and hard difficulties before starting a match. The selected difficulty correctly loads the corresponding AI controller. The selection is clearly visible to the player. The game starts with the chosen difficulty applied.	Gameplay	4	3	2	Gabrielle

Sprint 2 Test Cases

No.	User Stories	Acceptance Criteria	Story Points		
US_03	As a player, I want to be able to challenge an easy AI opponent so that I can practice basic gameplay mechanics.	An EasyController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack at random. The AI moves in the general direction of the player. The AI has a slower reaction update time, making its behavior less responsive and easier to predict.	10		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_03_01	Easy AI Controller Loads	Launch the game and start a match selecting easy difficulty. Observe the AI behavior at the start of the match.	The EasyController is active and controls the AI opponent.	3	
TC_03_02	Easy AI Performs Random Actions	Start a match on easy difficulty and observe the AI over time.	The AI performs actions such as block, jump, move, and attack in a random and simple manner.	3	
TC_03_03	Easy AI Moves Toward Player	Start a match on easy difficulty and position the player away from the AI. Observe AI movement.	The AI moves generally in the direction of the player.	2	
TC_03_04	Easy AI Has Slow Reaction Time	Start a match on easy difficulty and perform actions near the AI. Observe how quickly the AI responds.	The AI responds slowly, making it easier for the player to react and counter.	2	
US_04	As a player, I want to be able to challenge a medium AI opponent so that I can experience a balanced level of difficulty.	A MediumController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack with more consistency than random behavior. The AI moves toward the player more effectively than the easy AI. The AI has a moderate reaction update time, making it more responsive than easy but not overly difficult.	10		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_04_01	Medium AI Controller Loads	Launch the game and start a match selecting medium difficulty. Observe the AI behavior at the start of the match.	The MediumController is active and controls the AI opponent.	3	
TC_04_02	Medium AI Performs More Consistent Actions	Start a match on medium difficulty and observe the AI over time.	The AI performs actions more consistently and less randomly than easy difficulty.	3	
TC_04_03	Medium AI Tracks Player Movement	Start a match on medium difficulty and move the player around the stage. Observe AI movement.	The AI follows and positions itself toward the player more effectively than easy AI.	2	
TC_04_04	Medium AI Has Moderate Reaction Time	Start a match on medium difficulty and perform actions near the AI. Observe response timing.	The AI responds faster than easy difficulty but not instantly.	2	

Sprint 2 Test Cases Continued

No.	User Stories	Acceptance Criteria	Story Points		
US_05	As a player, I want to be able to challenge a hard AI opponent so that I can face a highly challenging and intelligent opponent.	A HardController script is implemented using machine learning techniques. Unity ML-Agents are installed and configured with a working Python environment. The AI overrides CollectObservations, OnActionReceived, and OnEpisodeBegin methods. A reward and penalty system is implemented to guide AI learning. The AI is trained using a Python training process. The resulting trained neural network model is imported and used in the game. The hard AI demonstrates advanced and responsive behavior compared to easy and medium difficulties.	15		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_05_01	Hard AI Controller and Model Load	Launch the game and start a match selecting hard difficulty. Observe the AI at the start of the match.	The HardController and trained neural network model are loaded and active.	3	
TC_05_02	Hard AI Demonstrates Advanced Behavior	Start a match on hard difficulty and observe the AI during gameplay.	The AI shows intelligent behavior such as well-timed attacks, blocking, and movement.	5	
TC_05_03	Hard AI Reacts Quickly to Player Actions	Start a match on hard difficulty and perform various actions near the AI.	The AI responds quickly and adapts to player actions with minimal delay.	2	
TC_05_04	Hard AI Training Integration Works	Launch the game after importing the trained model and play a match on hard difficulty.	The AI behavior reflects the trained model and runs without errors or missing functionality.	5	
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	6 (2 already earned in sprint 1)		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_11_01	Camera Shows Both Fighters	Start a match and observe camera positioning during gameplay.	Both fighters remain visible on screen at the same time.	3	
TC_11_02	Camera Follows Fighter Movement	Move fighters across the stage to opposite ends.	Camera adjusts position to keep both fighters visible.	3	

Sprint 2 Test Cases Continued

No.	User Stories	Acceptance Criteria	Story Points		
US_14	As a player, I want to be able to select between multiple difficulties so that I can choose a gameplay experience that matches my skill level.	The game provides an option to select between easy, medium, and hard difficulties before starting a match. The selected difficulty correctly loads the corresponding AI controller. The selection is clearly visible to the player. The game starts with the chosen difficulty applied.	4		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_14_01	Difficulty Selection Menu Displays	Launch the game and navigate to the mode or match setup screen.	The player is presented with options to select easy, medium, or hard difficulty.	2	
TC_14_02	Selected Difficulty Persists Into Match	Launch the game, select any difficulty, and start a match. Observe gameplay settings.	The match runs using the selected difficulty without reverting or changing unexpectedly.	1	
TC_14_03	Difficulty Selection Loads Correct AI	Launch the game, select difficulty, and start a match. Observe AI behavior.	The AI behaves according to easy, medium, and hard difficulty.	2	

Sprint 2 Stories Completed

No.	User Stories	Acceptance Criteria	Story Points		
US_03	As a player, I want to be able to challenge an easy AI opponent so that I can practice basic gameplay mechanics.	An EasyController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack at random. The AI moves in the general direction of the player. The AI has a slower reaction update time, making its behavior less responsive and easier to predict.	10		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_03_01	Easy AI Controller Loads	Launch the game and start a match selecting easy difficulty. Observe the AI behavior at the start of the match.	The EasyController is active and controls the AI opponent.	3	COMPLETE
TC_03_02	Easy AI Performs Random Actions	Start a match on easy difficulty and observe the AI over time.	The AI performs actions such as block, jump, move, and attack in a random and simple manner.	3	COMPLETE
TC_03_03	Easy AI Moves Toward Player	Start a match on easy difficulty and position the player away from the AI. Observe AI movement.	The AI moves generally in the direction of the player.	2	COMPLETE
TC_03_04	Easy AI Has Slow Reaction Time	Start a match on easy difficulty and perform actions near the AI. Observe how quickly the AI responds.	The AI responds slowly, making it easier for the player to react and counter.	2	COMPLETE
US_04	As a player, I want to be able to challenge a medium AI opponent so that I can experience a balanced level of difficulty.	A MediumController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack with more consistency than random behavior. The AI moves toward the player more effectively than the easy AI. The AI has a moderate reaction update time, making it more responsive than easy but not overly difficult.	10		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_04_01	Medium AI Controller Loads	Launch the game and start a match selecting medium difficulty. Observe the AI behavior at the start of the match.	The MediumController is active and controls the AI opponent.	3	COMPLETE
TC_04_02	Medium AI Performs More Consistent Actions	Start a match on medium difficulty and observe the AI over time.	The AI performs actions more consistently and less randomly than easy difficulty.	3	COMPLETE
TC_04_03	Medium AI Tracks Player Movement	Start a match on medium difficulty and move the player around the stage. Observe AI movement.	The AI follows and positions itself toward the player more effectively than easy AI.	2	COMPLETE
TC_04_04	Medium AI Has Moderate Reaction Time	Start a match on medium difficulty and perform actions near the AI. Observe response timing.	The AI responds faster than easy difficulty but not instantly.	2	COMPLETE
US_05	As a player, I want to be able to challenge a hard AI opponent so that I can face a highly challenging and intelligent opponent.	A HardController script is implemented using machine learning techniques. Unity ML Agents are installed and configured with a working Python environment. The AI overrides CollectObservations, OnActionReceived, and OnEpisodeBegin methods. A reward and penalty system is implemented to guide AI learning. The AI is trained using a Python training process. The resulting trained neural network model is imported and used in the game. The hard AI demonstrates advanced and responsive behavior compared to easy and medium difficulties.	15		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_05_01	Hard AI Controller and Model Load	Launch the game and start a match selecting hard difficulty. Observe the AI at the start of the match.	The HardController and trained neural network model are loaded and active.	3	COMPLETE
TC_05_02	Hard AI Demonstrates Advanced Behavior	Start a match on hard difficulty and observe the AI during gameplay.	The AI shows intelligent behavior such as well timed attacks, blocking, and movement.	5	COMPLETE
TC_05_03	Hard AI Reacts Quickly to Player Actions	Start a match on hard difficulty and perform various actions near the AI.	The AI responds quickly and adapts to player actions with minimal delay.	2	COMPLETE
TC_05_04	Hard AI Training Integration Works	Launch the game after importing the trained model and play a match on hard difficulty.	The AI behavior reflects the trained model and runs without errors or missing functionality.	5	COMPLETE

Sprint 2 Stories Not Completed

No.	User Stories	Acceptance Criteria	Story Points		
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	6 (2 already earned in sprint 1)		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_11_01	Camera Shows Both Fighters	Start a match and observe camera positioning during gameplay.	Both fighters remain visible on screen at the same time.	3	SPRINT 3
TC_11_02	Camera Follows Fighter Movement	Move fighters across the stage to opposite ends.	Camera adjusts position to keep both fighters visible.	3	SPRINT 3
US_14	As a player, I want to be able to select between multiple difficulties so that I can choose a gameplay experience that matches my skill level.	The game provides an option to select between easy, medium, and hard difficulties before starting a match. The selected difficulty correctly loads the corresponding AI controller. The selection is clearly visible to the player. The game starts with the chosen difficulty applied.	4		
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status
TC_14_01	Difficulty Selection Menu Displays	Launch the game and navigate to the mode or match setup screen.	The player is presented with options to select easy, medium, or hard difficulty.	2	SPRINT 3
TC_14_02	Selected Difficulty Persists Into Match	Launch the game, select any difficulty, and start a match. Observe gameplay settings.	The match runs using the selected difficulty without reverting or changing unexpectedly.	1	SPRINT 3
TC_14_03	Difficulty Selection Loads Correct AI	Launch the game, select difficulty, and start a match. Observe AI behavior.	The AI behaves according to easy, medium, and hard difficulty.	2	SPRINT 3

Stories Planned for Sprint 3

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint	Team Member
US_2	As a player, I want to design my character so that I can play with a customized fighter.	Player can choose character name. Player can design character appearance. Character preview is displayed before confirmation. User confirms selection to proceed. Selected character loads into the match	Character System	5	2	3	Johnny
US_6	As a player, I want the player and AI characters to use pixel art sprites so that the characters match the retro visual style.	Player character uses a pixel art sprite. AI opponent uses a pixel art sprite. Sprites replace any placeholder shapes for characters and display correctly during gameplay.	Visual Design	3	3	3	Shariah
US_12	As a player, I want the game environment to use pixel art graphics so that the stage matches the retro aesthetic.	The game background uses pixel art design. Background assets load correctly during gameplay. The environment visuals are consistent with the pixel art style and no placeholder backgrounds appear in the Sprint demo.	Visual Design	3	3	3	Shariah
US_13	As a player, I want character sprites to animate during movement and attacks so that the game feels dynamic and visually engaging.	Player sprites animate during movement and attacks. AI sprites animate during movement or attacks. Animations display smoothly during gameplay.	Visual Design	3	3	3	Shariah

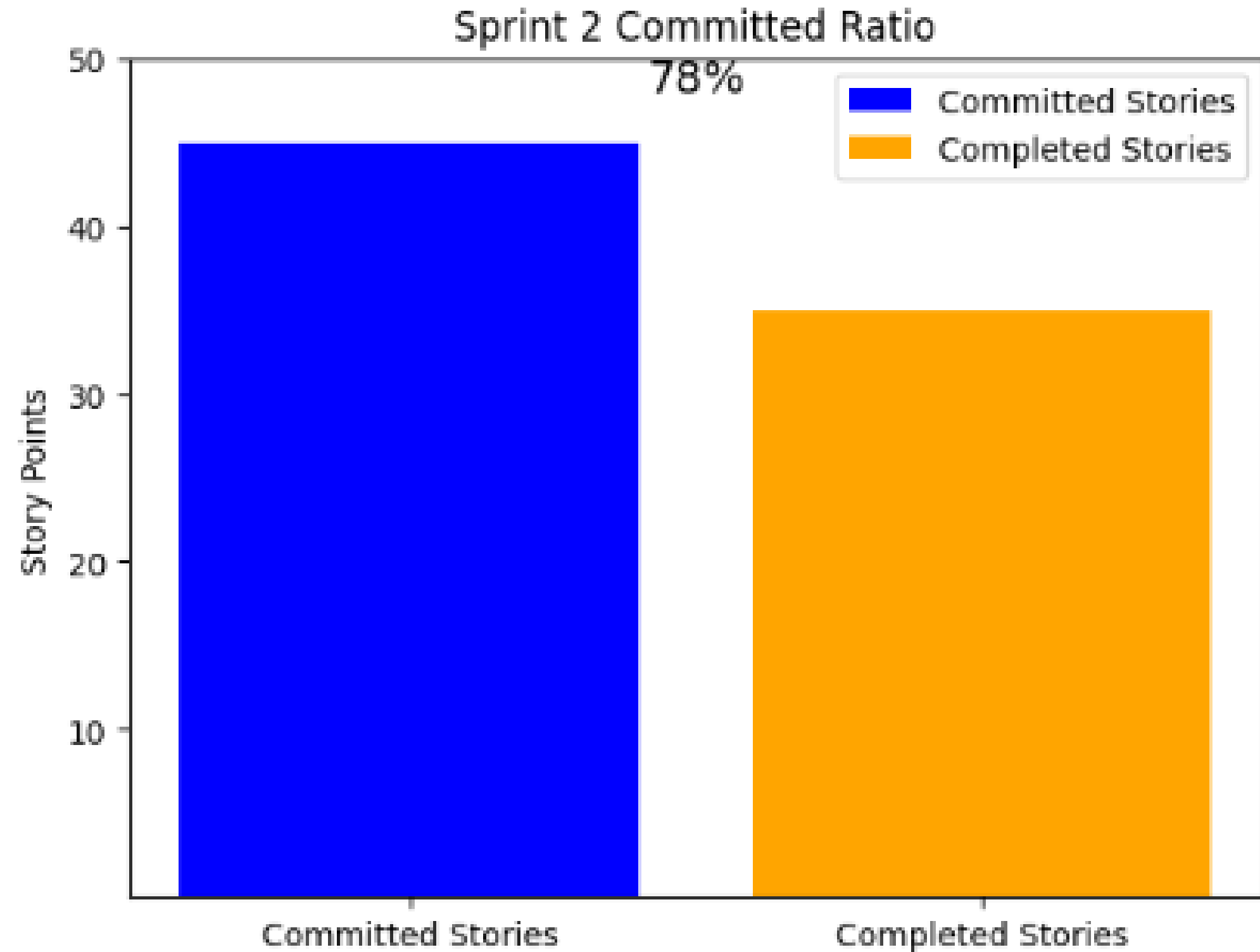
Stories Planned for Sprint 3

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint	Team Member
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	Camera System	6 (2 already earned in sprint 1)	3	2	Anthony
US_14	As a player, I want to be able to select between multiple difficulties so that I can choose a gameplay experience that matches my skill level.	The game provides an option to select between easy, medium, and hard difficulties before starting a match. The selected difficulty correctly loads the corresponding AI controller. The selection is clearly visible to the player. The game starts with the chosen difficulty applied.	Gameplay	4	3	2	Gabrielle

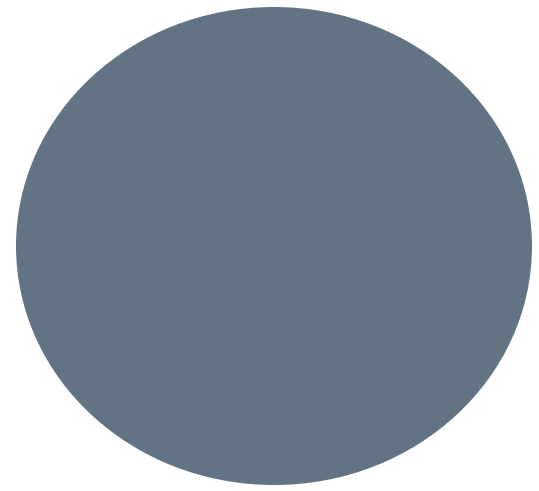
Team Velocity – Sprint 2

Story Points Committed: 45

Story Points Completed: 35



Team Velocity



Historical ratios:

- Sprint 1: 70.73%
- Sprint 2: 77.78%

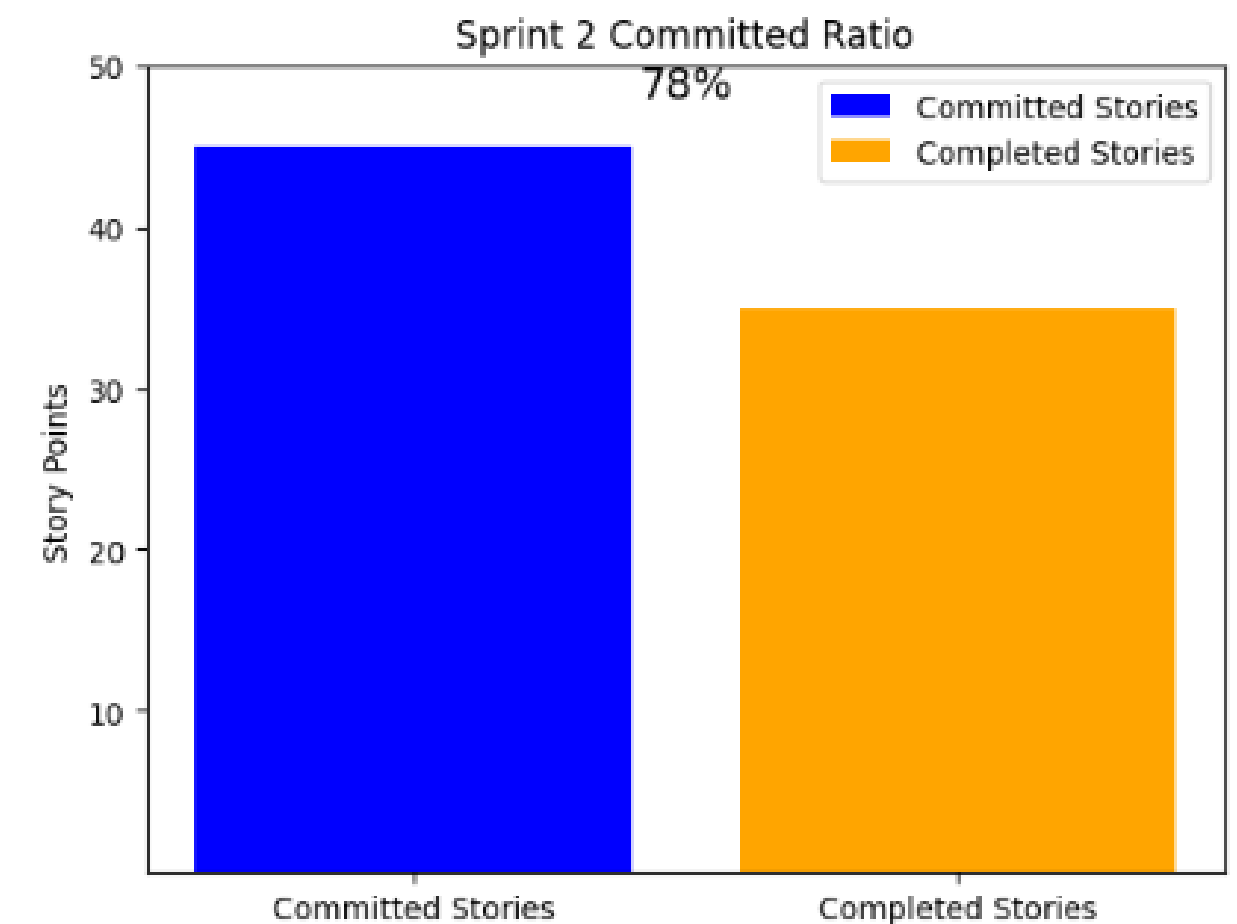
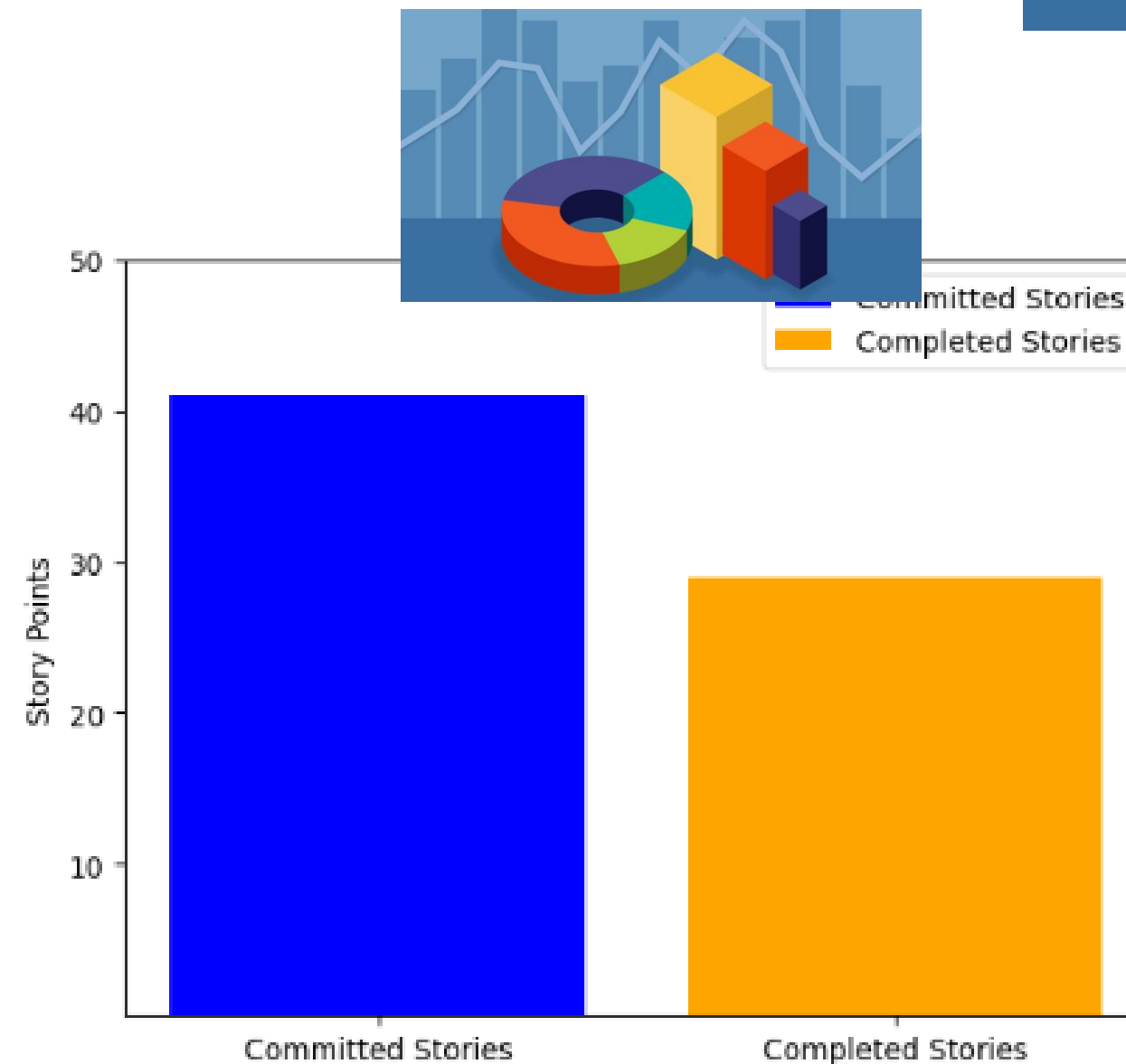


Improvement:

- +7.05%

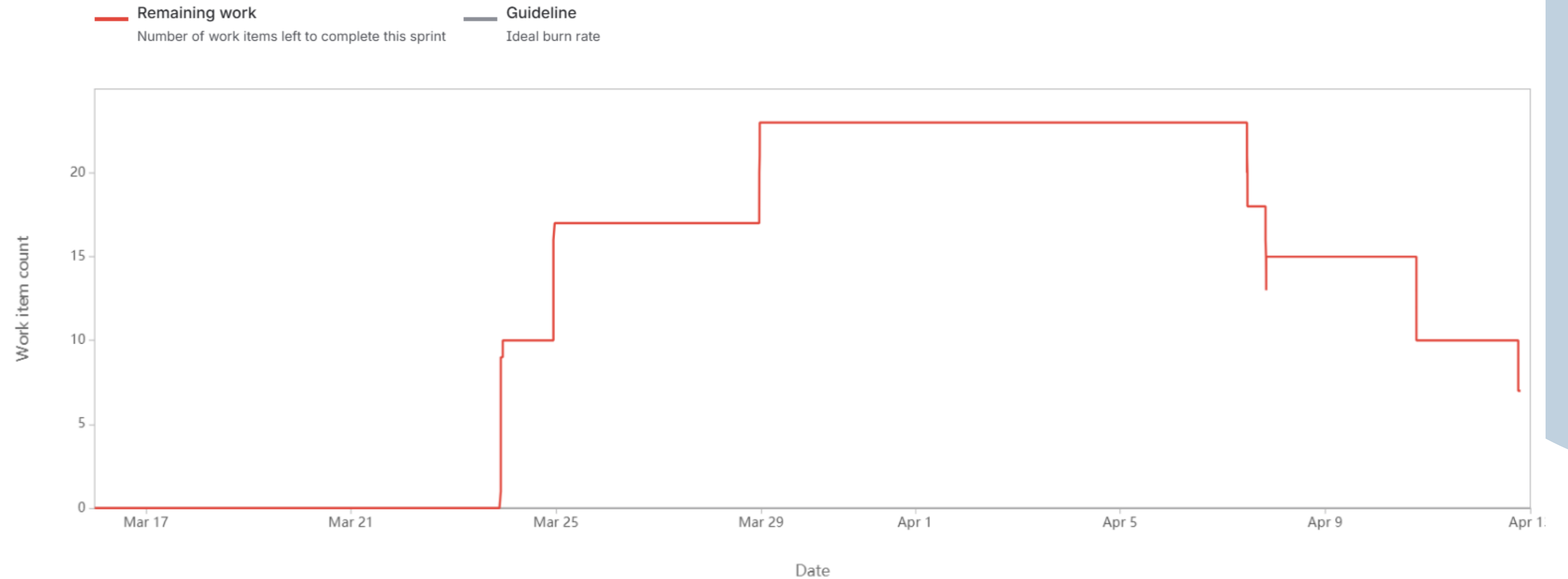
Average ratio:

- 74.26%



Burndown Chart

Date - March 16th, 2026 - April 13th, 2026



Report: SCRUM Sprint 2

*Work item added after sprint 1

Retrospectives

QuestKeeper Retrospective Sprint 2

What went well +

The technical side was smoother and the code was better polished. + 0	Communication was great + 0
Test cases and user stories were completed well. + 0	We received positive feedback from the professor for our work in the last sprint. + 0
All roles were divided among the team members. + 0	

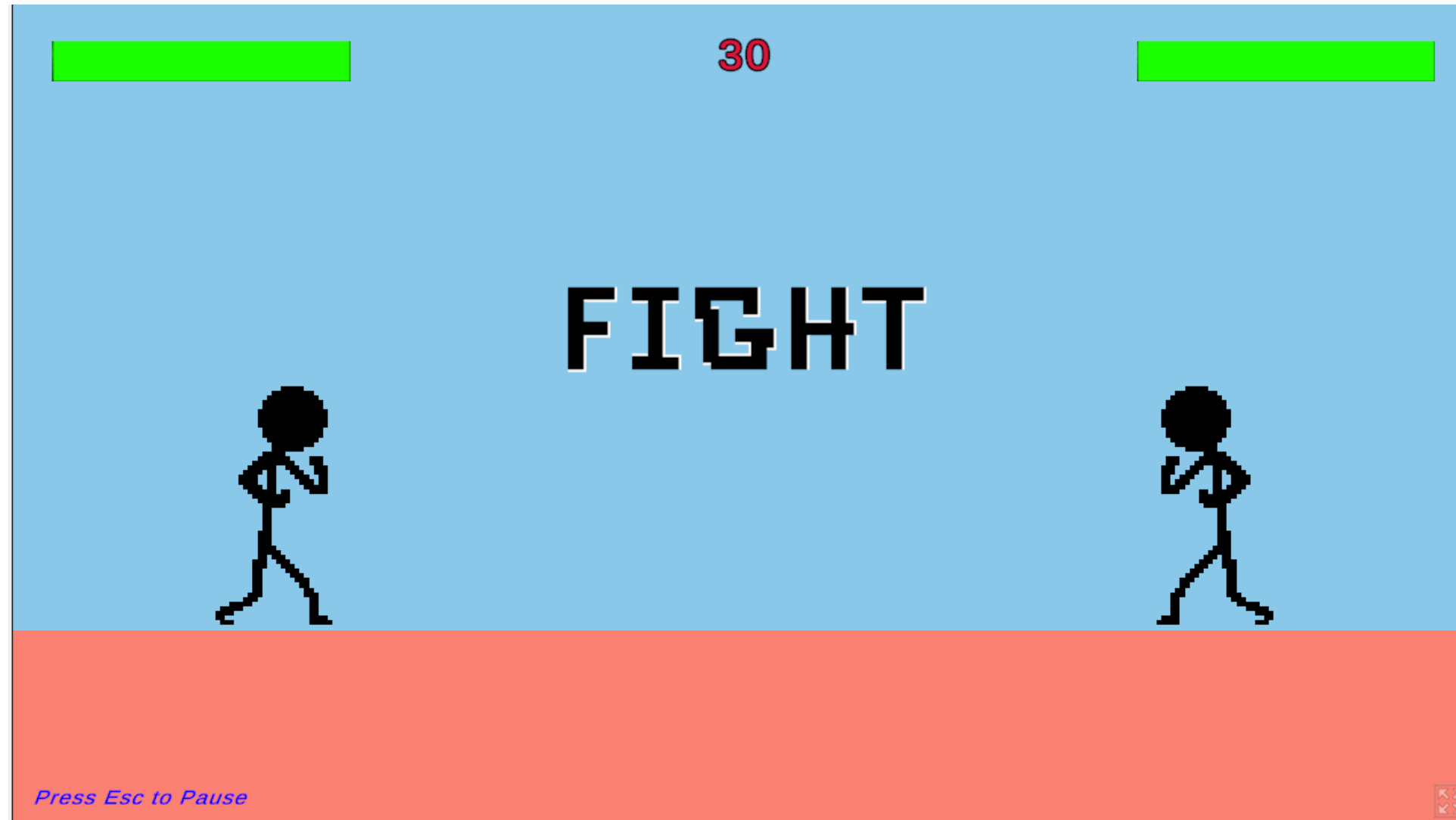
What can be improved +

Version control (GitHub): pull from the main branch every day to prevent losing updates and push changes frequently. + 0	Pay more attention to the current script. + 0
Work together more on the technical documentation. + 0	Split the work into small features. + 0
Add the professor to our team to look more professional. + 0	

Action Items +

More communication about technical aspects. Ask more about current tasks and any problems + 0	Check the created code to build the script in the same manner. + 0
Do more work in Jira: open tasks and mark what we finish for now + 0	Improve collaboration to reduce stress. + 0
Keep time to polish the final result. + 0	

Project Demo



Adaptive Arena Demo



Wiki Page Link

<https://github.com/htmhw/2026SA-QuestKeeper/wiki>





**Thank
You**